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EE6222 Machine Vision

Video (Part II)

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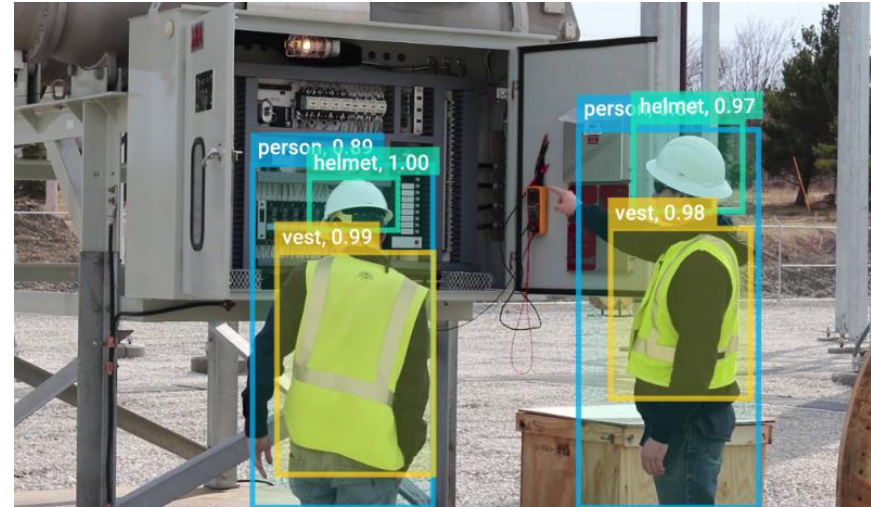
Major Topics

- Object Detection & Tracking
- Action Recognition
- Video Event/Anomaly Detection
- Video Enhancement
- Optical Flow

Video Analysis Applications



Counting people in shopping mall



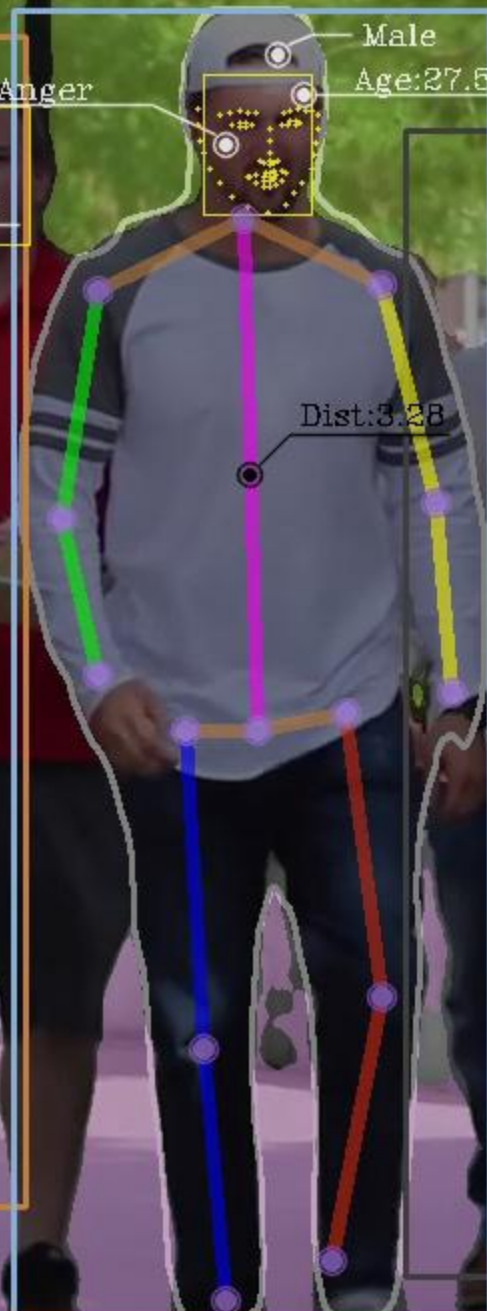
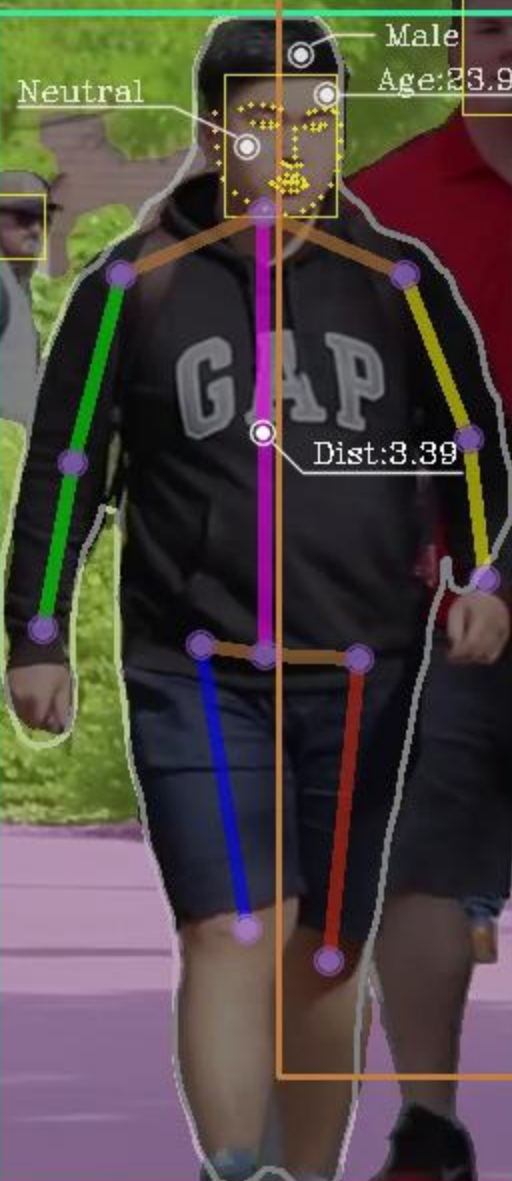
Inspection in power and utility industry



Fall detection



Action recognition: stretching leg



Green: Tree
Purple: Road
Gray: Building

Big Daws

Video

What we see:



What a computer sees:



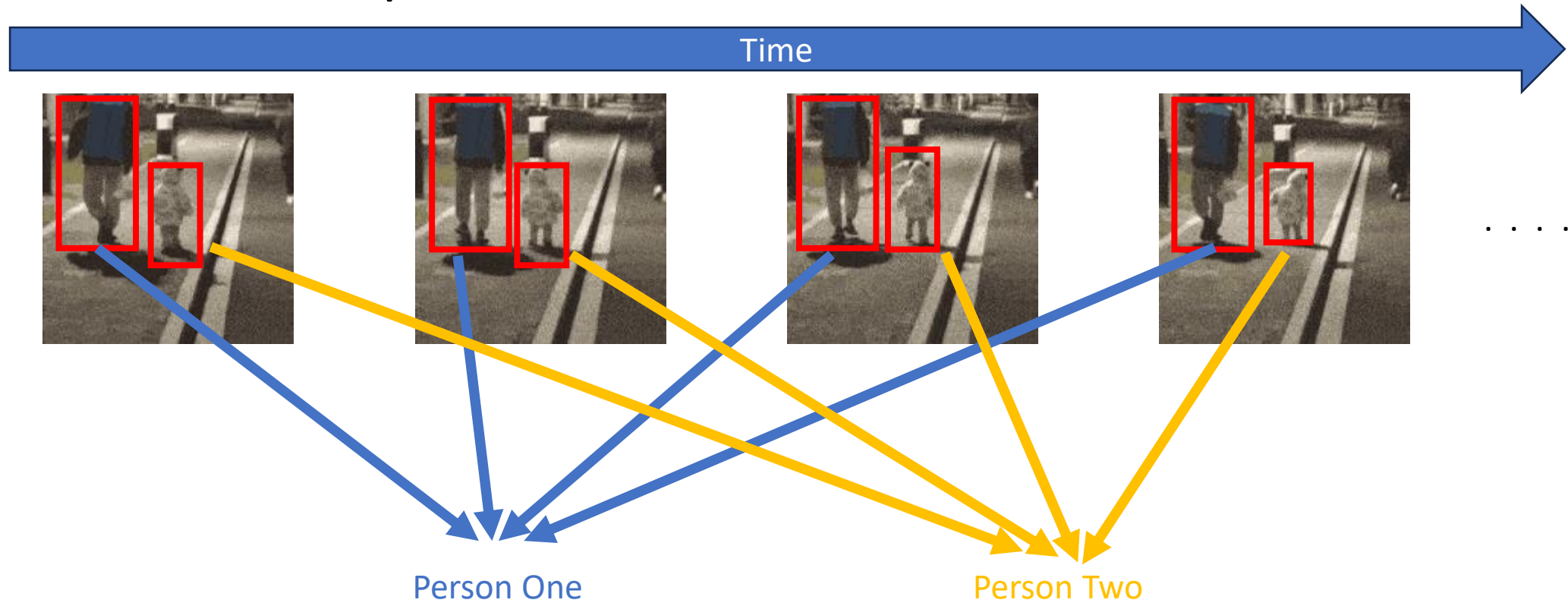
...

Time

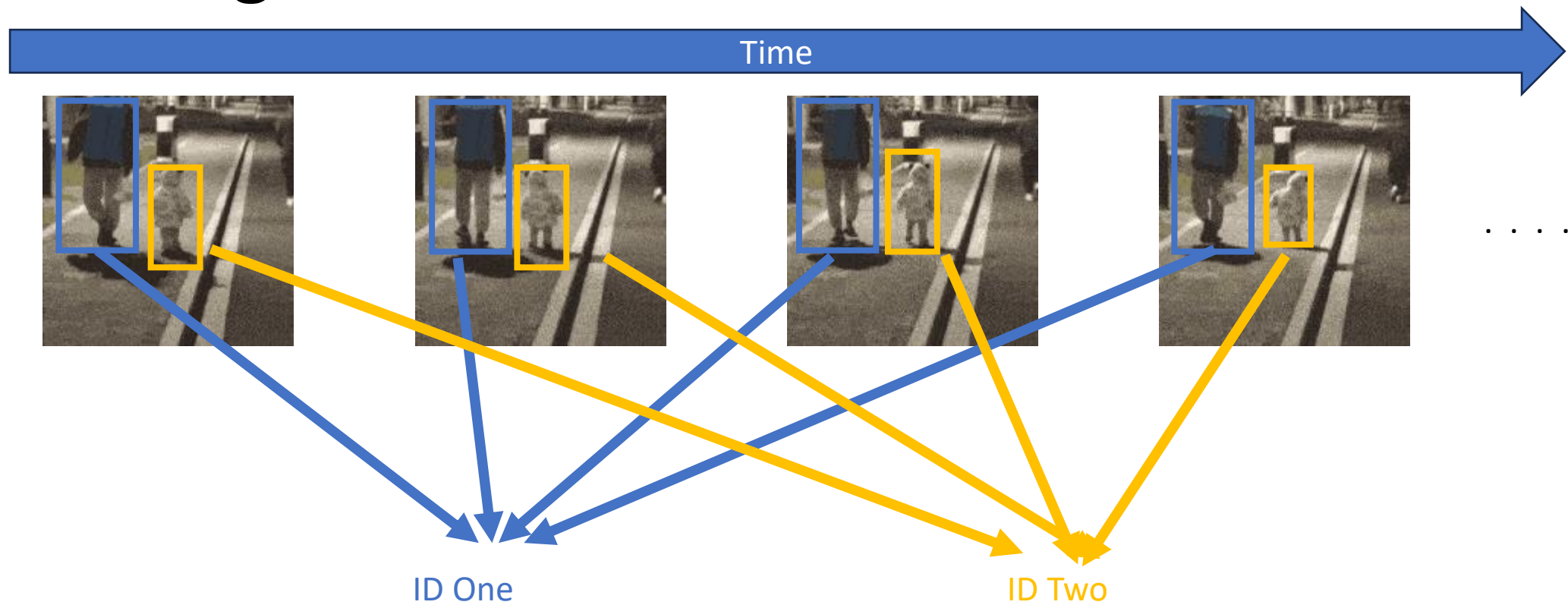
Video is a sequence of images with correlations among the images.

Detection and Tracking

How a computer understands the video



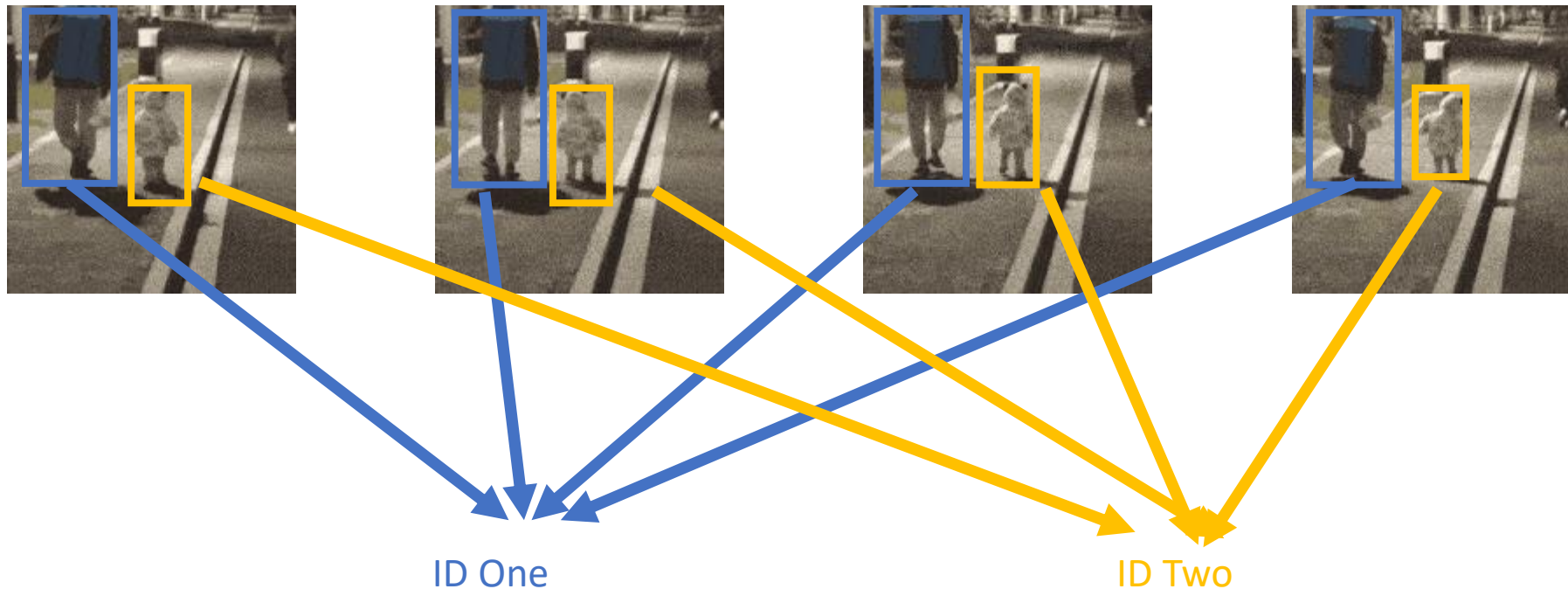
Tracking

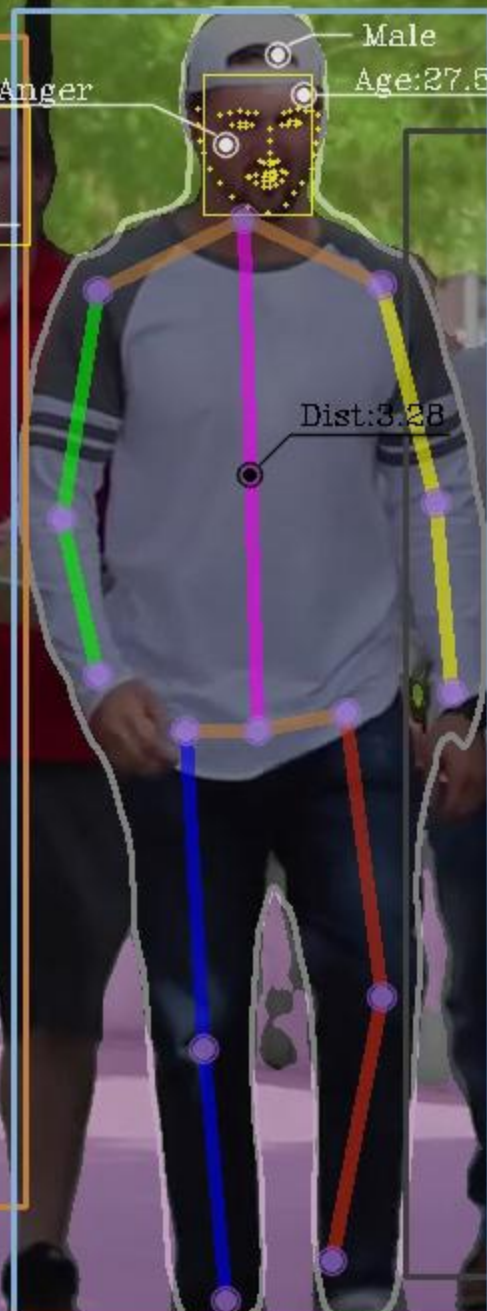
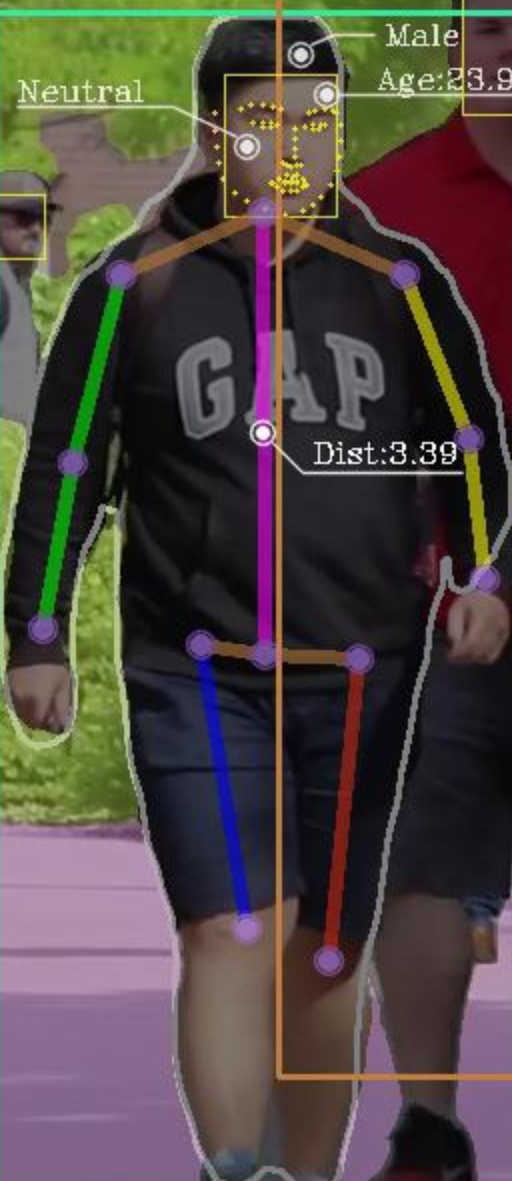


Object Detection & Tracking

Typical way:

1. Detect/segment objects
2. Associate detections over time





Green: Tree
Purple: Road
Gray: Building

Raw Video Source: goo.gl/LqyMQM
Big Dawgs

Main Difference between Video and Image

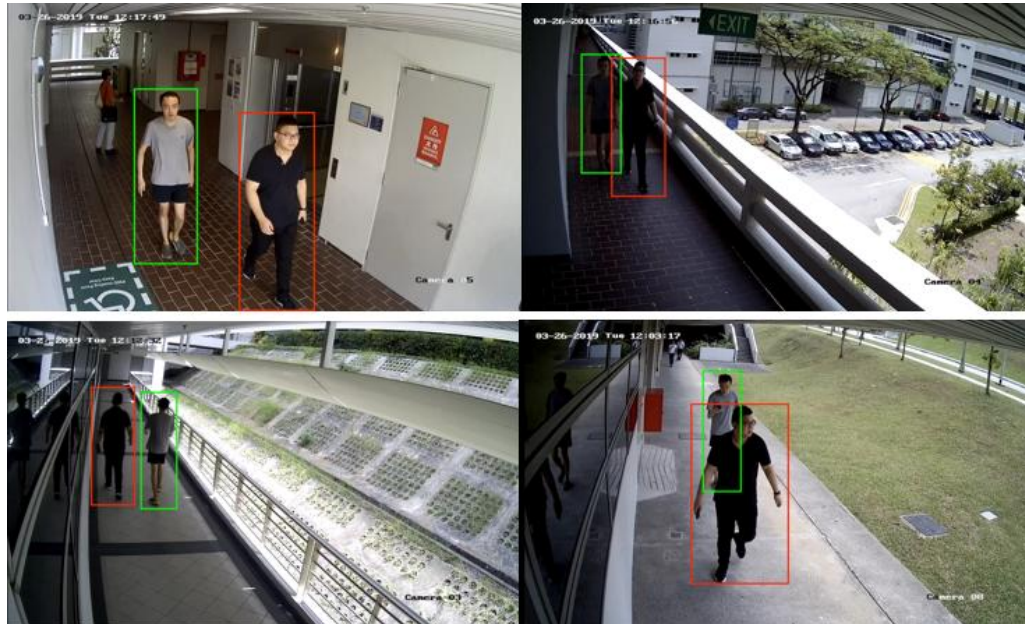
- Temporal Continuity

1. Image object detection: each image is independently processed
2. Video object detection: there is a continuity across time. Temporal information can be used to improve detection, tracking etc.

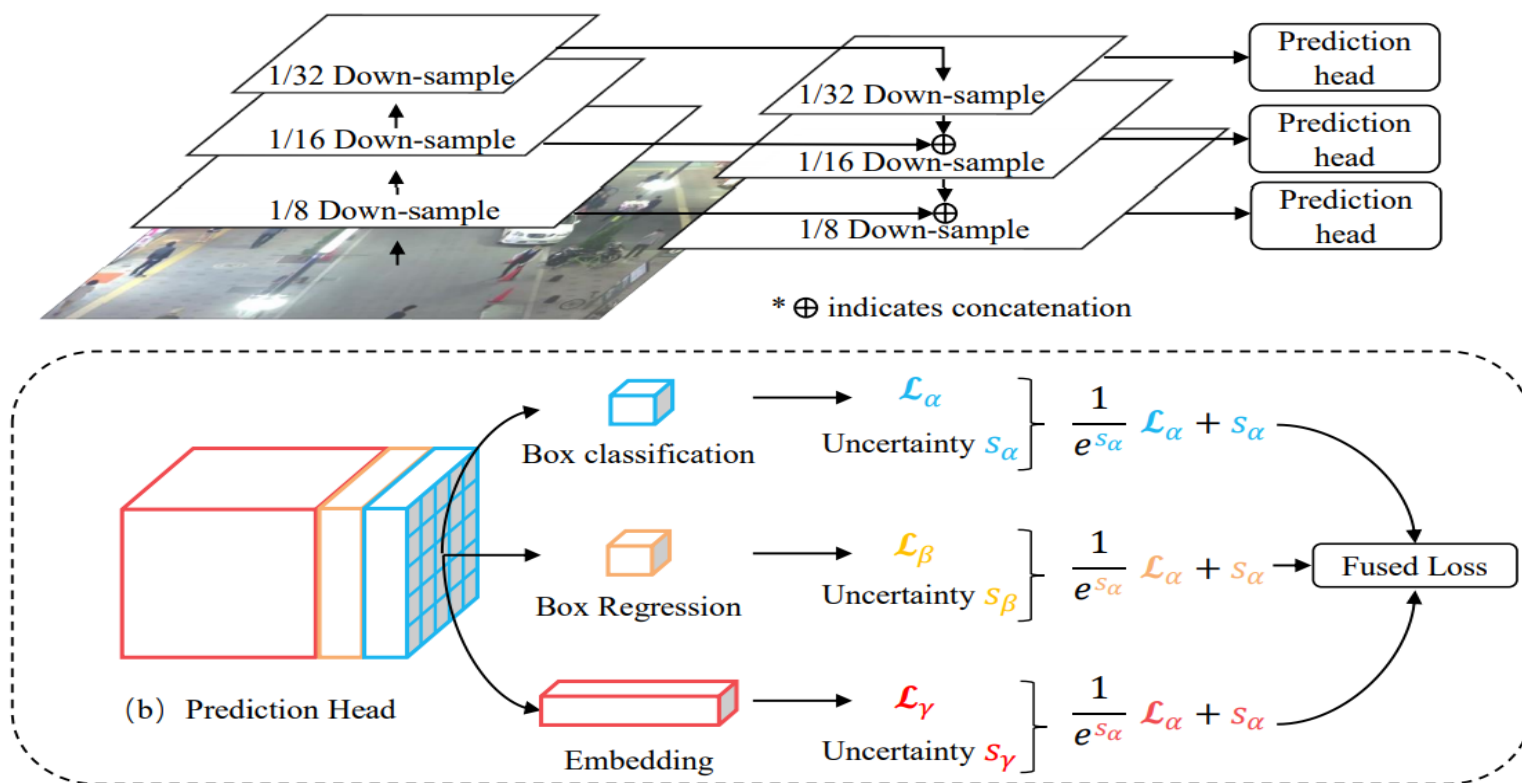


Re-ID

- Re-identification (Re-ID for short) is defined as the problem of matching object/people across disjoint camera views in a multi-camera system.
- It is often achieved by detection and tracking



Joint Detection and Embedding (JDE)



ReID vs. JDE

- Dji ReID



FairMOT JDE



What are the differences?

Video(Action, Event) Recognition/Detection

<http://www.thumos.info/home.html>



Video Events

Basic event or action detection



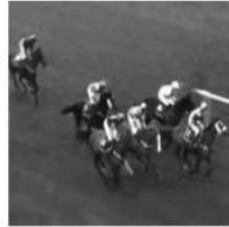
push



pushup



ride
bike



ride
horse



run



shake
hands



shoot
ball

Complex or high-level event detection



Wedding Ceremony

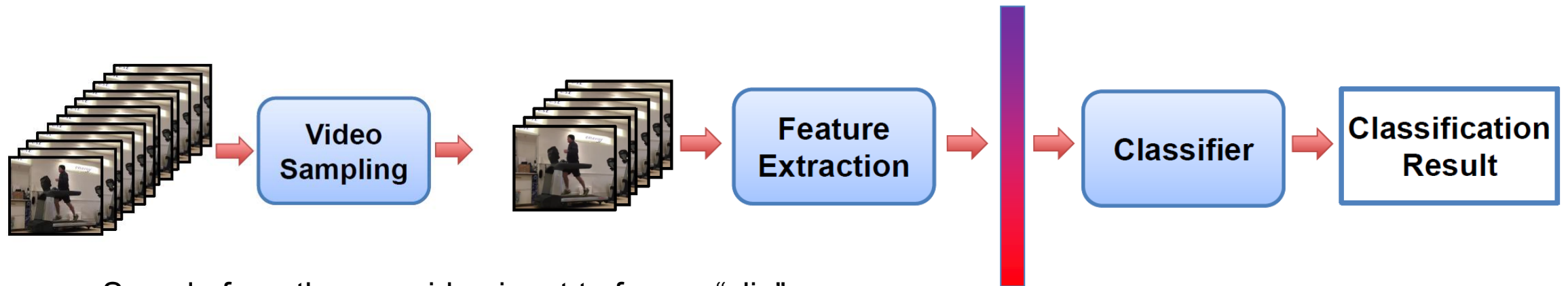
Video Event



How to detect “Goal”?

Action Recognition

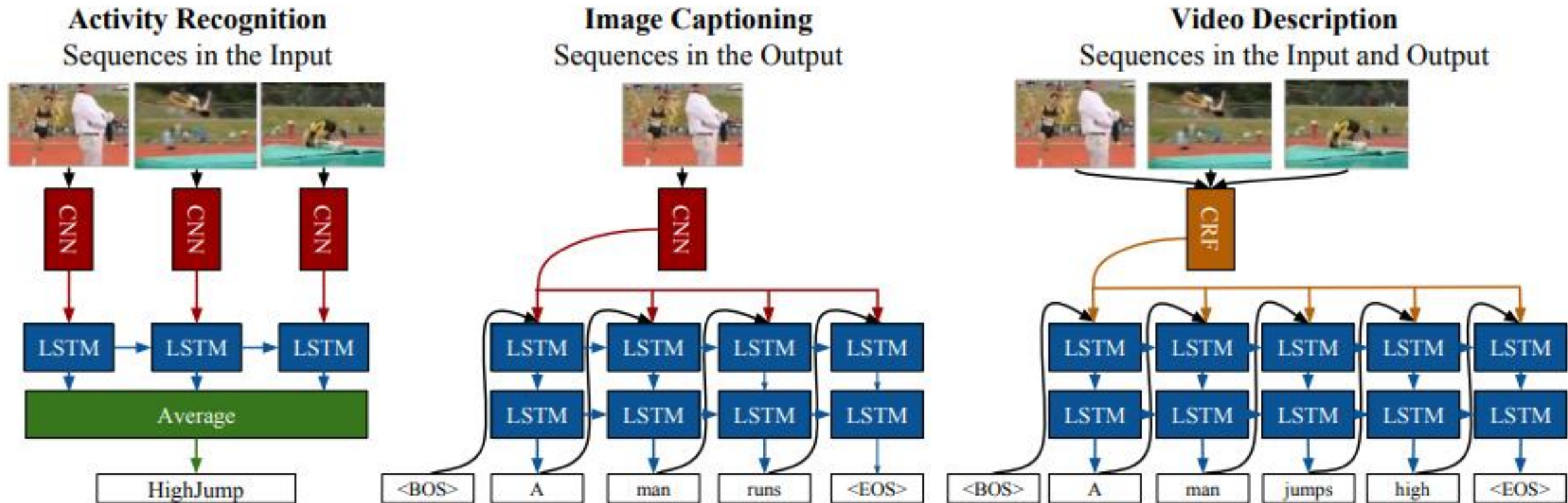
- Action Recognition is a computer vision task that involves recognizing human actions in videos or images.



Sample from the raw video input to form a “clip”.
Feature extraction from video clip for recognition.

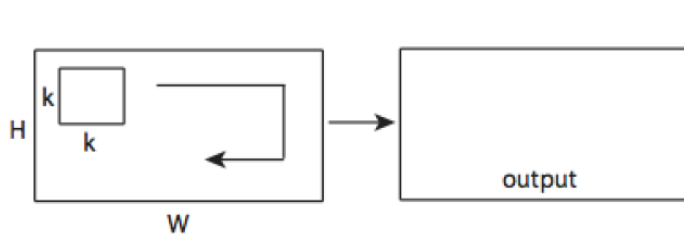
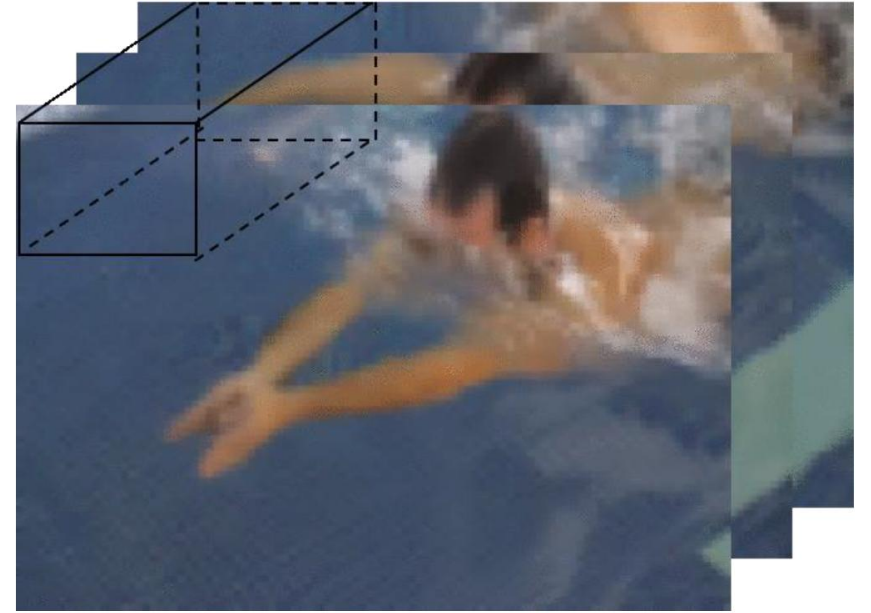
Action recognition is similar to object recognition.

Long-term Recurrent Convolutional Network

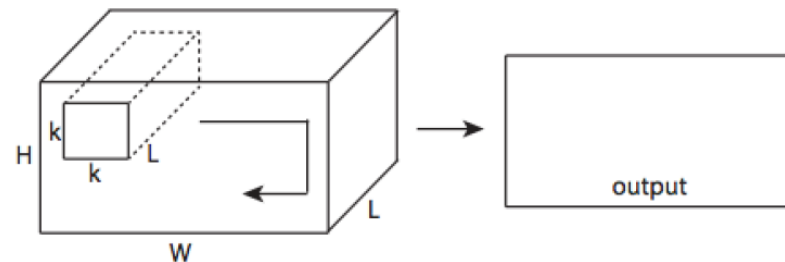


C3D: 3D Convolutional Networks

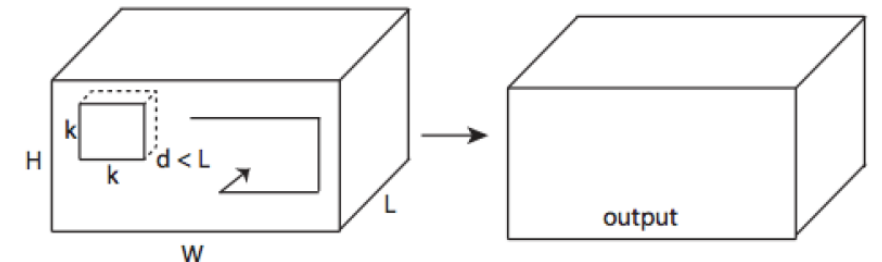
- Du Tran et al., “Learning Spatiotemporal Features with 3D Convolutional Networks,” 2014 ([ArxivLink](#))
- Repurposing 3D convolutional networks as feature extractors
- Extensive search for best 3D convolutional kernel and architecture
- Using deconvolutional layers to interpret model decision



(a) 2D convolution



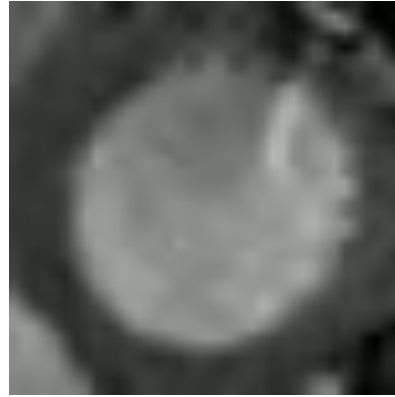
(b) 2D convolution on multiple frames



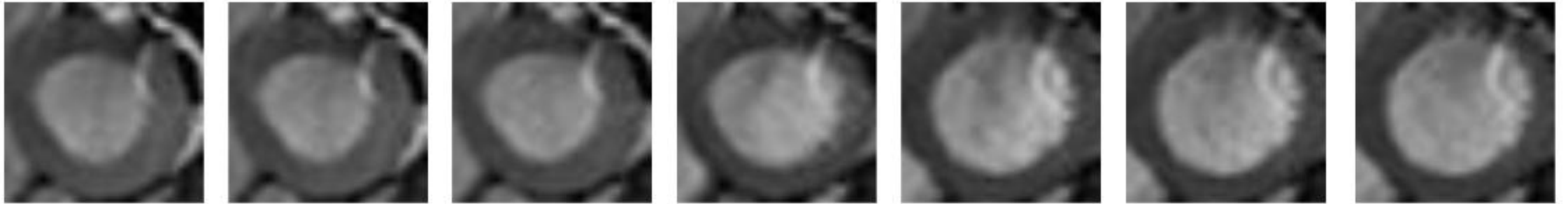
(c) 3D convolution

Cine MRI for Microvascular Obstruction Identification

- What we see:

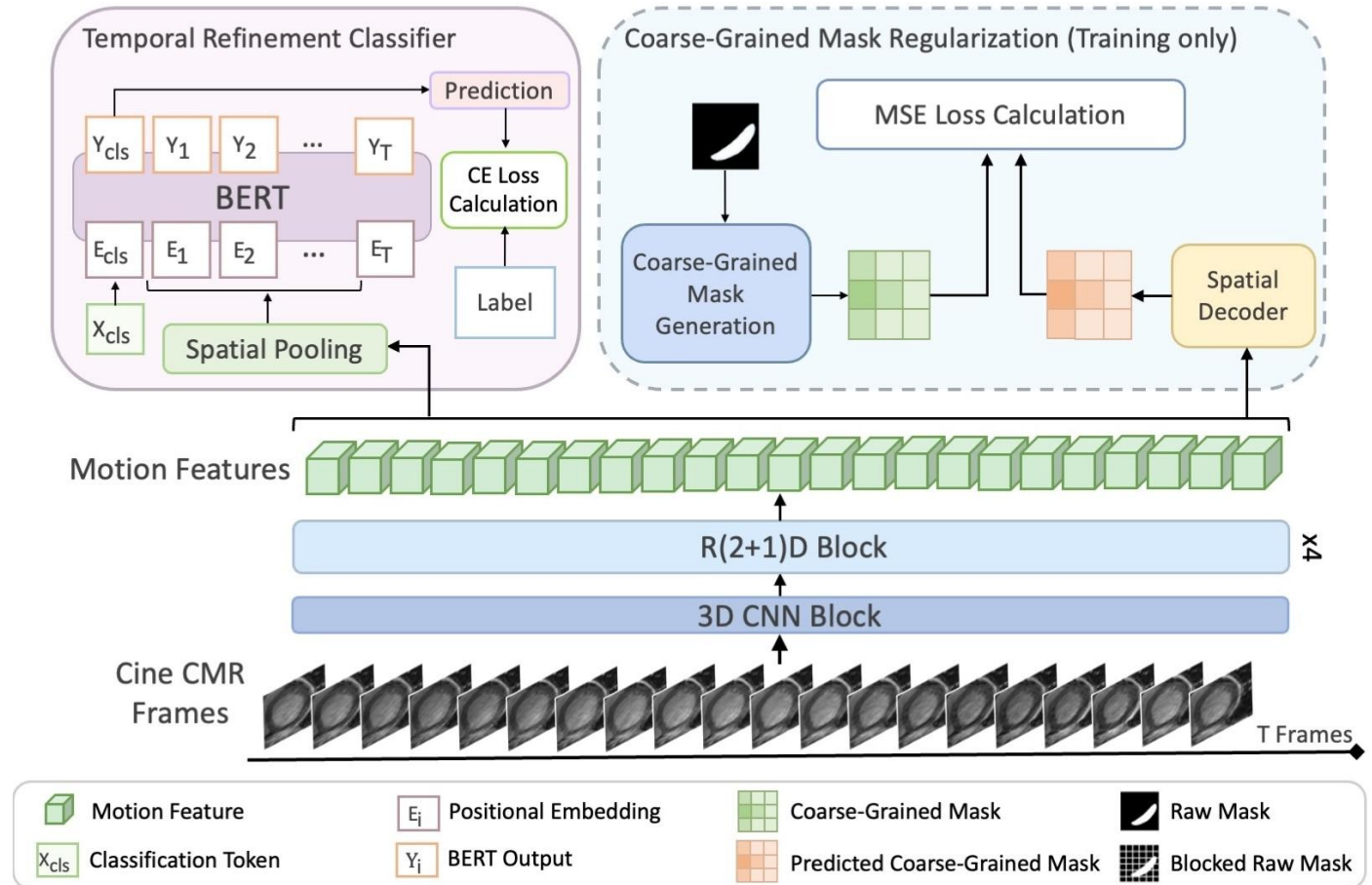


- What a computer sees:



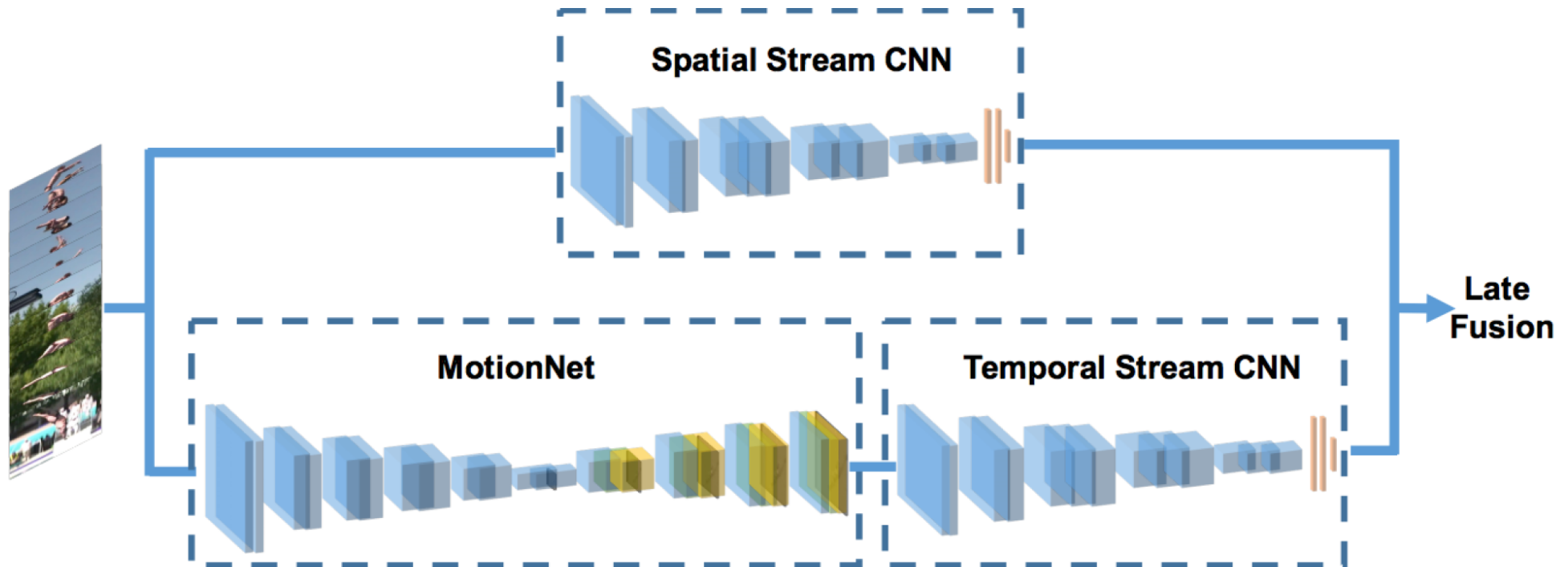
Healthy or Unhealthy?

Cine MRI for Microvascular Obstruction Identification



Hidden Two Stream

- Zhu et al., “Hidden Two-Stream Convolutional Networks for Action Recognition,” 2017
- •Novel architecture for generating optical flow input on-the-fly using a separate network



Video Enhancement

Motion blur is everywhere



Object Motion



Camera Motion

Blur model

Point-spread function

$$\begin{aligned} g(n_1, n_2) &= d(n_1, n_2) * f(n_1, n_2) + w(n_1, n_2) \\ &= \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} d(i, j) f(n_1 - i, n_2 - j) + w(n_1, n_2) \end{aligned}$$

- $d(x, y)$ takes on nonnegative values only, because of the physics of the underlying image formation process;
- when real-valued images are dealt with the point-spread function $d(x, y)$ is real-valued too;
- the imperfections in the image formation process are modeled as passive operations on the data, i.e, no “energy” is absorbed or generated.

$$\sum_{n_1=1}^{N-1} \sum_{n_2=1}^{M-1} d(n_1, n_2) = 1$$

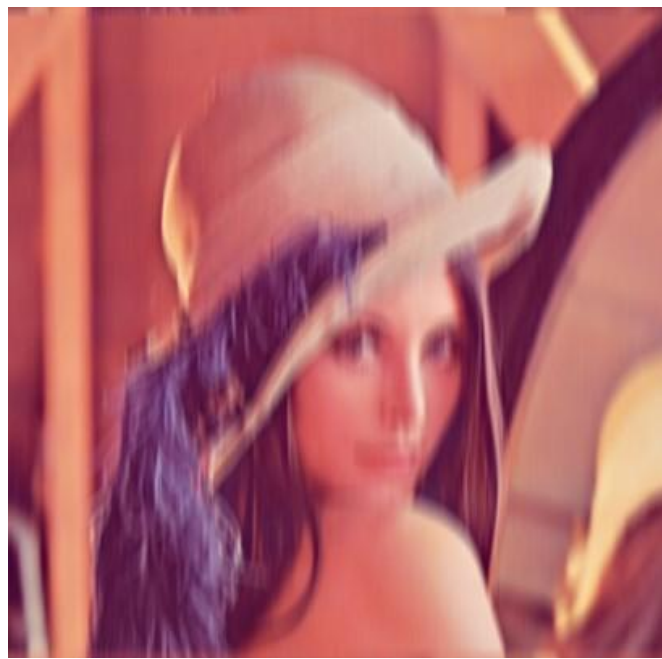
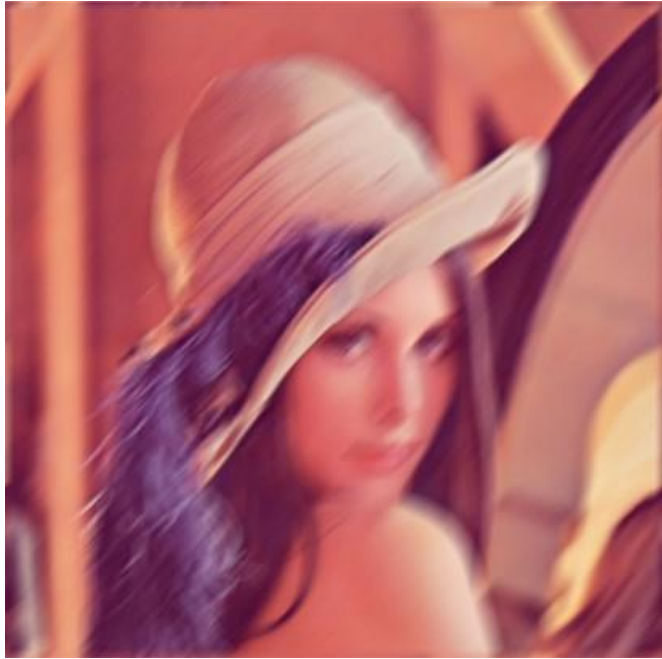
Blur model

No blur:

PSF is modeled as a unit pulse:

$$d(n_1, n_2) = \delta(n_1, n_2) = \begin{cases} 1 & \text{if } n_1 = n_2 = 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\begin{aligned} g(n_1, n_2) &= \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} d(i, j) f(n_1 - i, n_2 - j) + w(n_1, n_2) \\ &= f(n_1, n_2) + w(n_1, n_2) \end{aligned}$$



Blind Image Deconvolution



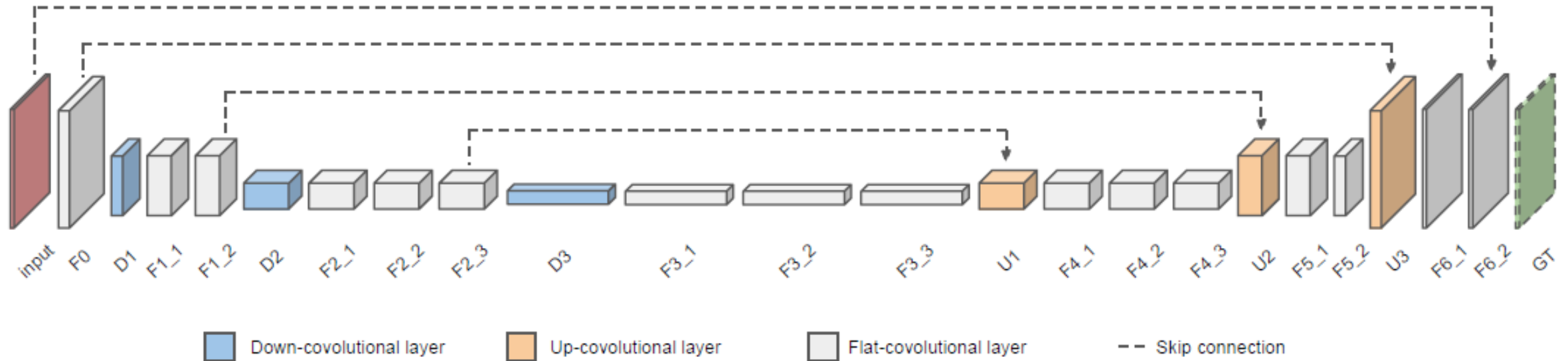
- Accurate Point Spread Function (PSF) Needed.

Deep video deblur

Input: the stacked nearby frames

Structure: Encoder – Decoder

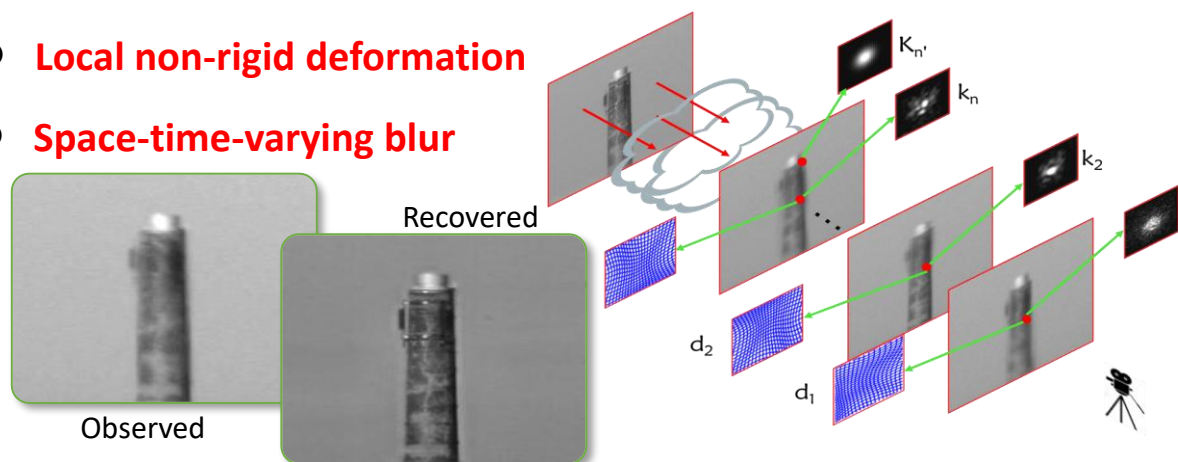
Project home: <https://www.cs.ubc.ca/labs/imager/tr/2017/DeepVideoDeblurring/>



S. Su, et al., "Deep Video Deblurring for Hand-Held Cameras," *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 237-246.

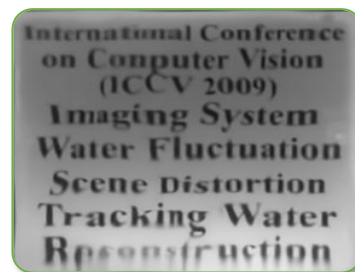
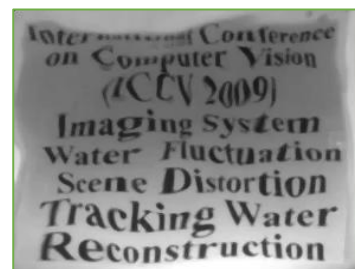
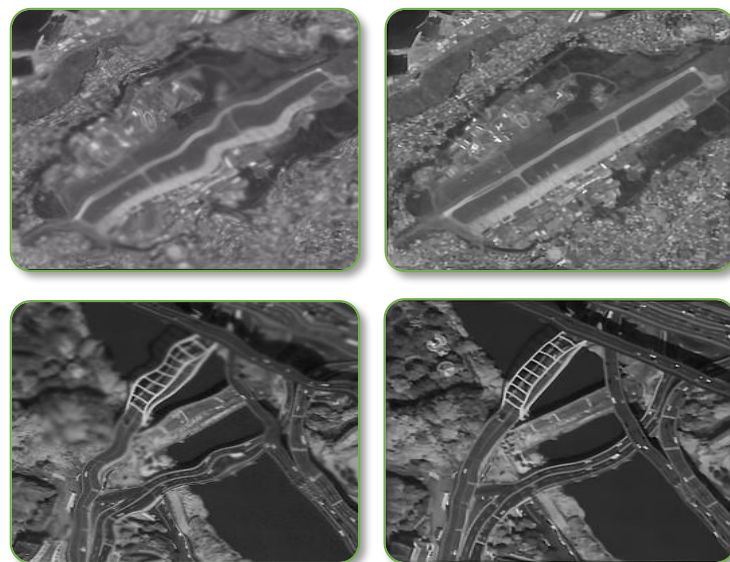
Video Turbulence Effect Remove

- **Local non-rigid deformation**
- **Space-time-varying blur**

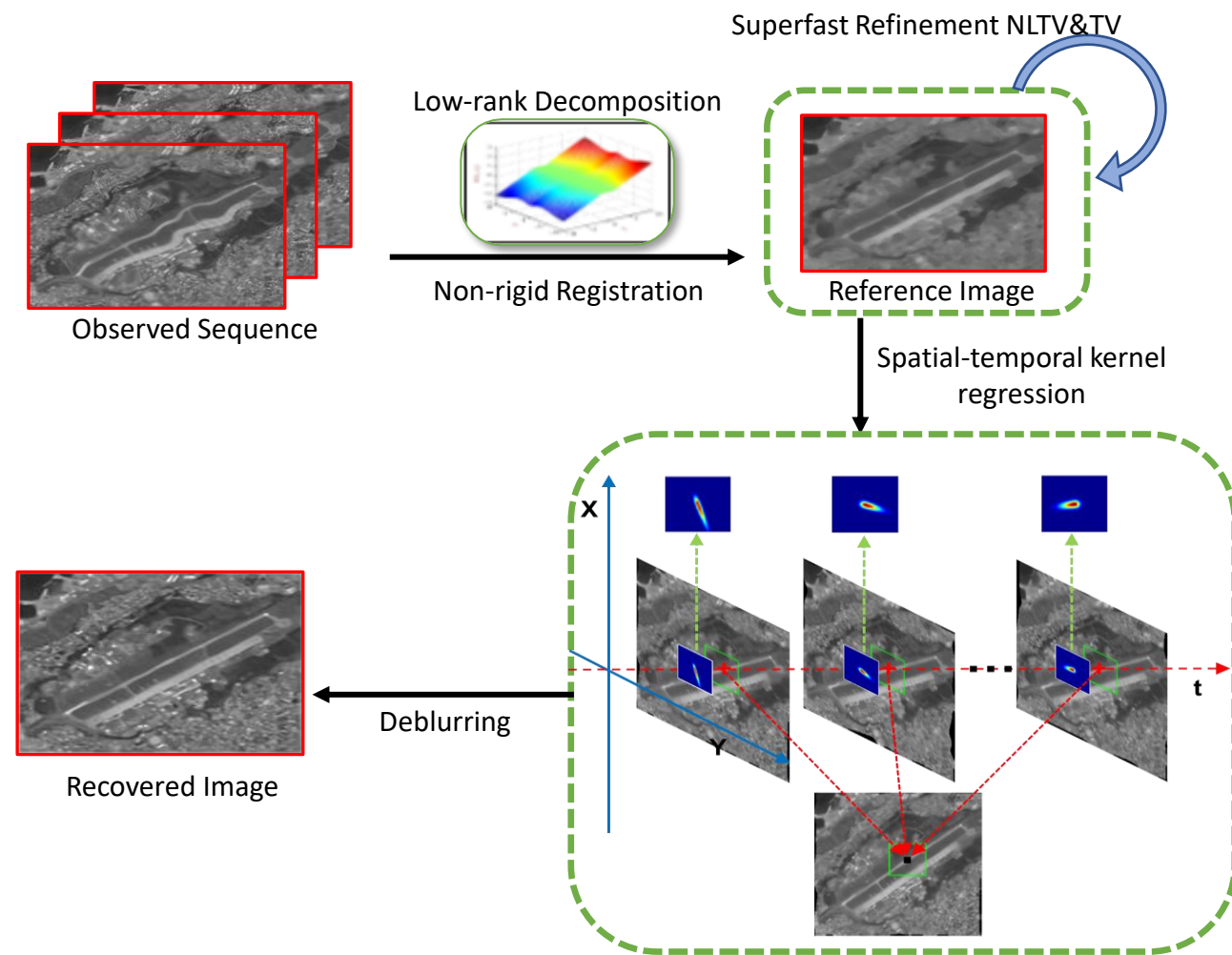


Air Turbulence

Water Turbulence



The Proposed Restoration Framework



Optical Flow

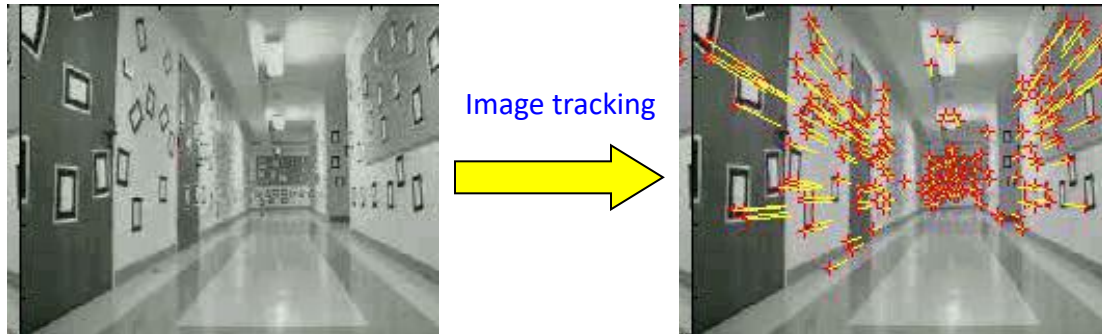
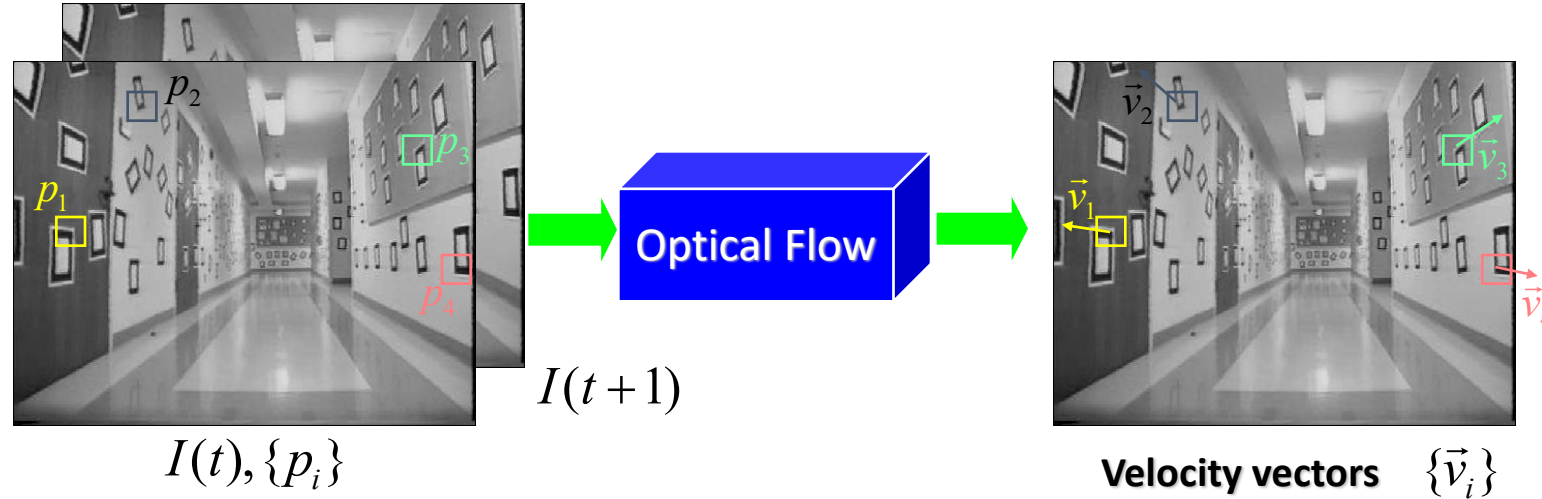


Image sequence
(single camera)

Tracked sequence

What is Optical Flow?

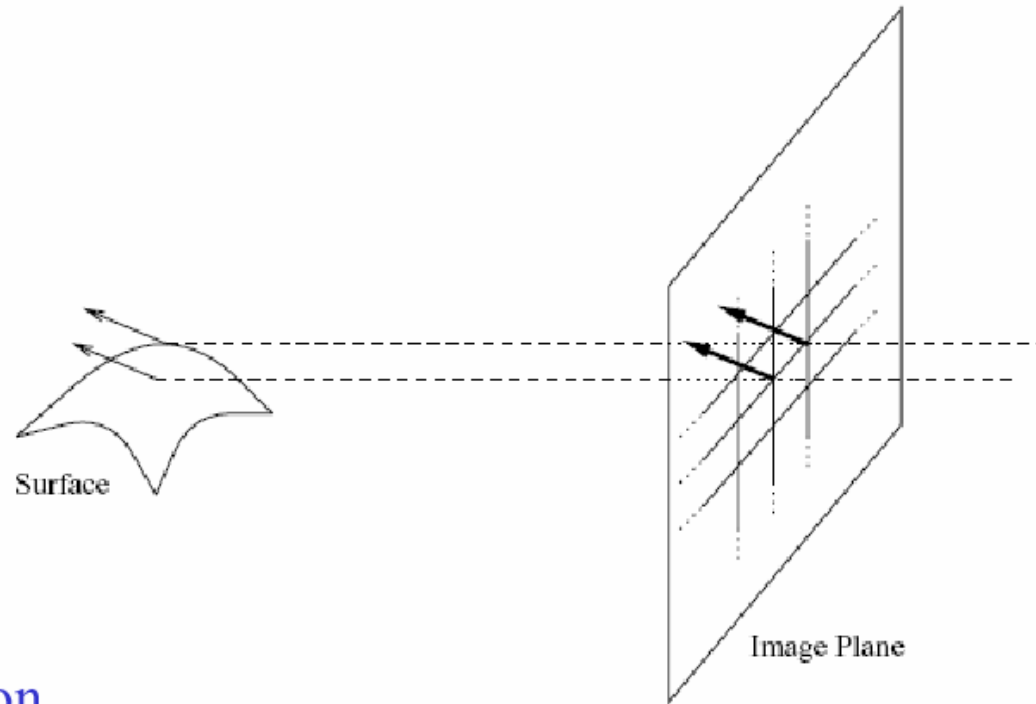


Optical flow is the relation of the motion field

- *the 2D projection of the physical movement of points relative to the observer to 2D displacement of pixel patches on the image plane.*

Optical Flow Assumptions:

Spatial Coherence

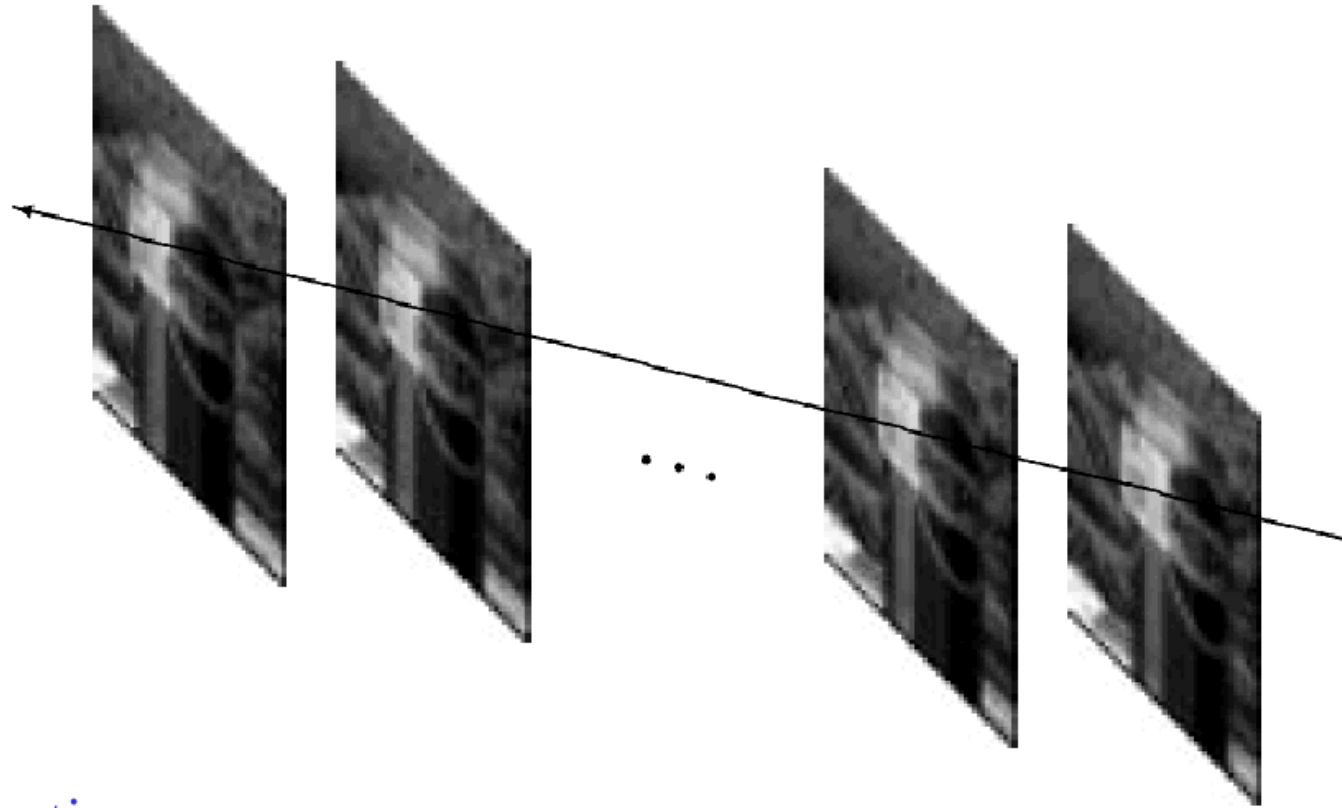


Assumption

- * Neighboring points in the scene typically belong to the same surface and hence typically have similar motions.
- * Since they also project to nearby points in the image, we expect spatial coherence in image flow.

Optical Flow Assumptions:

Temporal Persistence

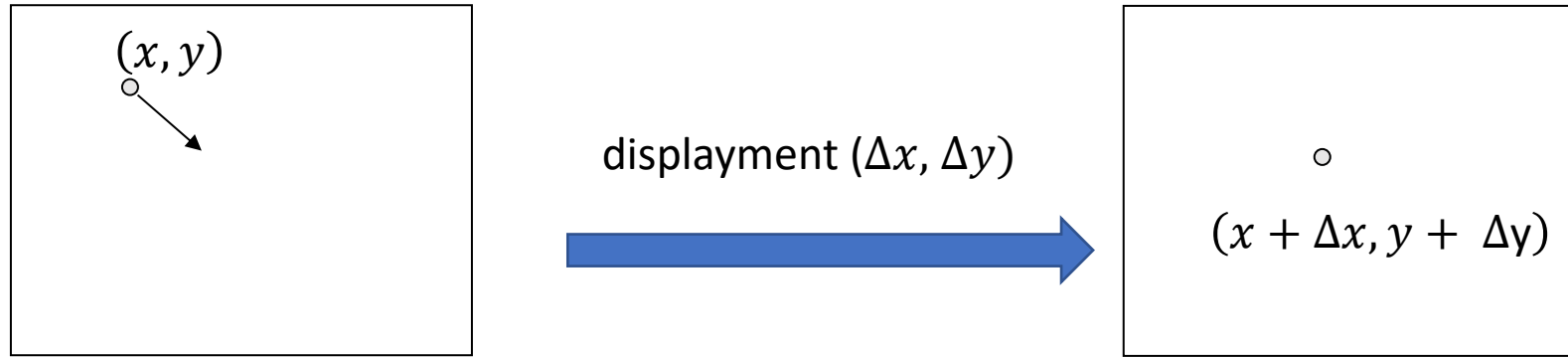


Assumption:

The image motion of a surface patch changes gradually over time.

Optical Flow Assumptions:

The brightness constancy constraint



Brightness Constancy Assumption

$$I(x, y, t) = I(x + \Delta x, y + \Delta y, t + \Delta t)$$

Brightness constancy equation

- Take Taylor expansion of $I(x + \Delta x, y + \Delta y, t + \Delta t)$ at (x, y, t) :

$$I(x + \Delta x, y + \Delta y, t + \Delta t) \approx I(x, y, t) + \frac{\partial I}{\partial x} \Delta x + \frac{\partial I}{\partial y} \Delta y + \frac{\partial I}{\partial t} \Delta t + \dots$$

Since $I(x + \Delta x, y + \Delta y, t + \Delta t) = I(x, y, t)$

$$\frac{\partial I}{\partial x} \Delta x + \frac{\partial I}{\partial y} \Delta y + \frac{\partial I}{\partial t} \Delta t \approx 0 \qquad \frac{\partial I}{\partial x} \frac{\Delta x}{\Delta t} + \frac{\partial I}{\partial y} \frac{\Delta y}{\Delta t} + \frac{\partial I}{\partial t} \approx 0$$

$$\frac{\partial I}{\partial x} \frac{dx}{dt} + \frac{\partial I}{\partial y} \frac{dy}{dt} + \frac{\partial I}{\partial t} = 0 \quad \text{or} \quad I_x \cdot \frac{dx}{dt} + I_y \cdot \frac{dy}{dt} + I_t = 0 \quad \text{or} \quad \nabla I \cdot \begin{bmatrix} \frac{dx}{dt} & \frac{dy}{dt} \end{bmatrix}^T + I_t = 0$$

The brightness constancy constraint

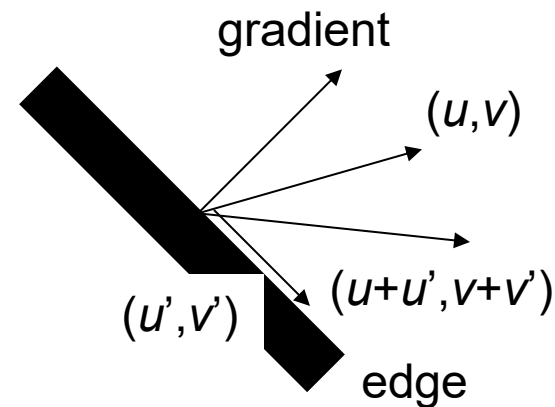
- How many equations and unknowns per pixel? $\nabla I \cdot [u \ v]^T + I_t = 0$

One equation (this is a scalar equation!), two unknowns (u,v)

The component of the motion perpendicular to the gradient (i.e., parallel to the edge) cannot be measured

If (u, v) satisfies the equation,
so does $(u+u', v+v')$ if

$$\nabla I \cdot [u' \ v']^T = 0$$



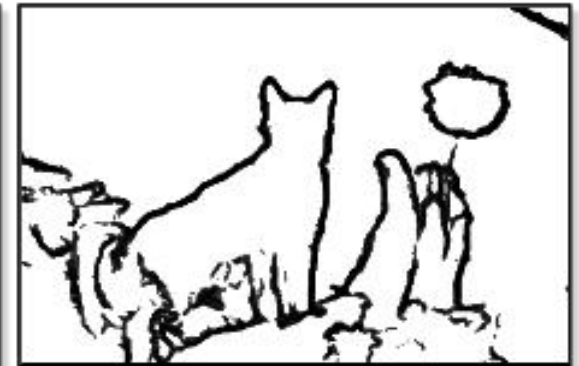
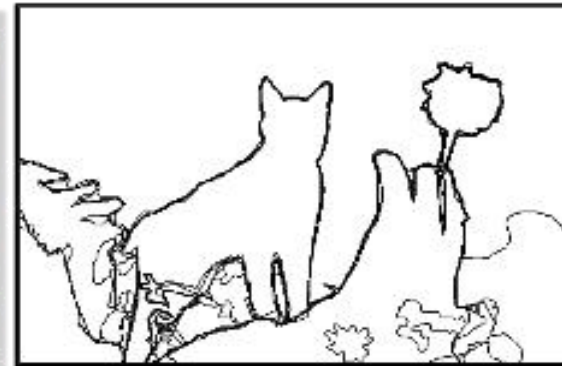
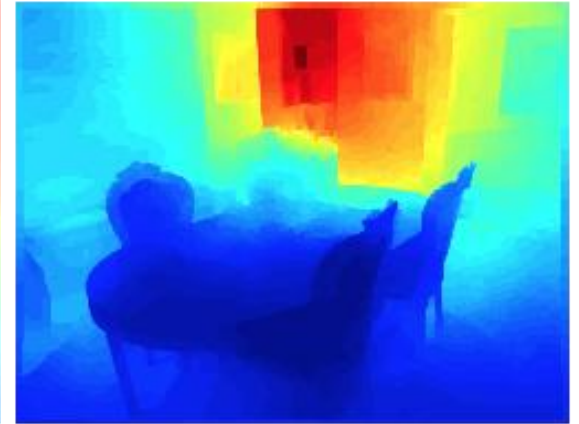
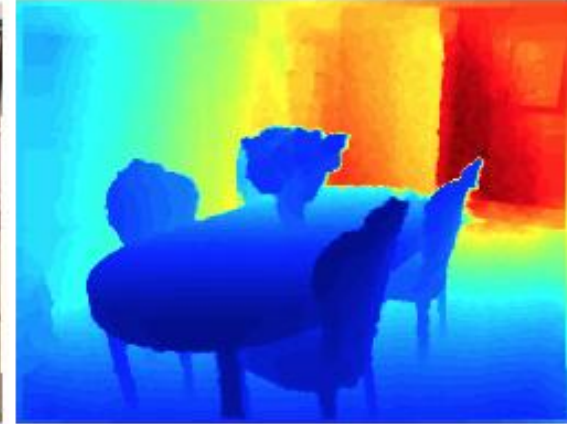
Video Segmentation

The process of segment objects from video sequences.

Segmentation: Pixels in, pixels out

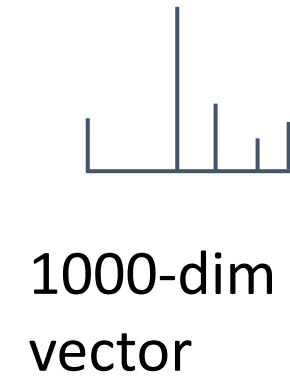
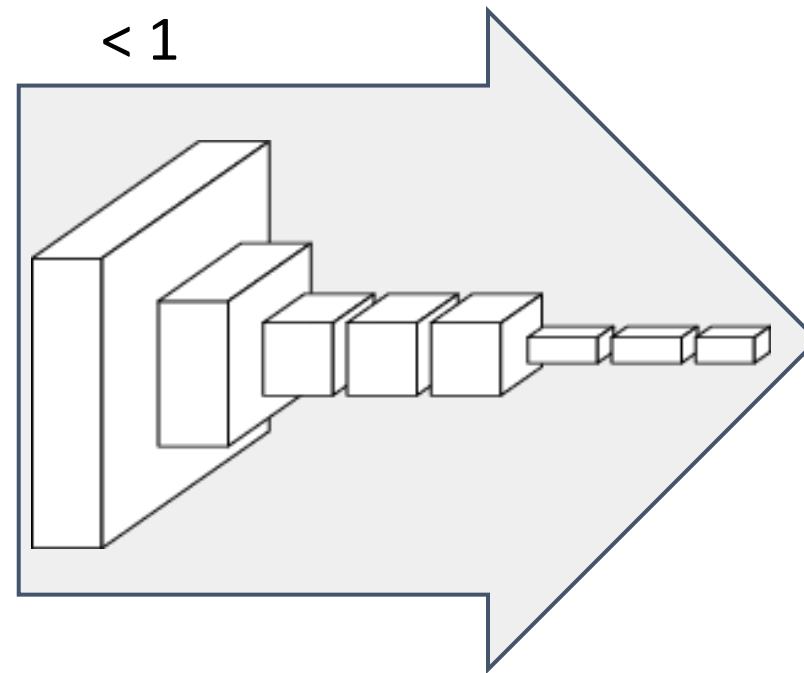
monocular depth estimation (Liu et al. 2015)

semantic
segmentation



boundary prediction (Xie & Tu 2015)

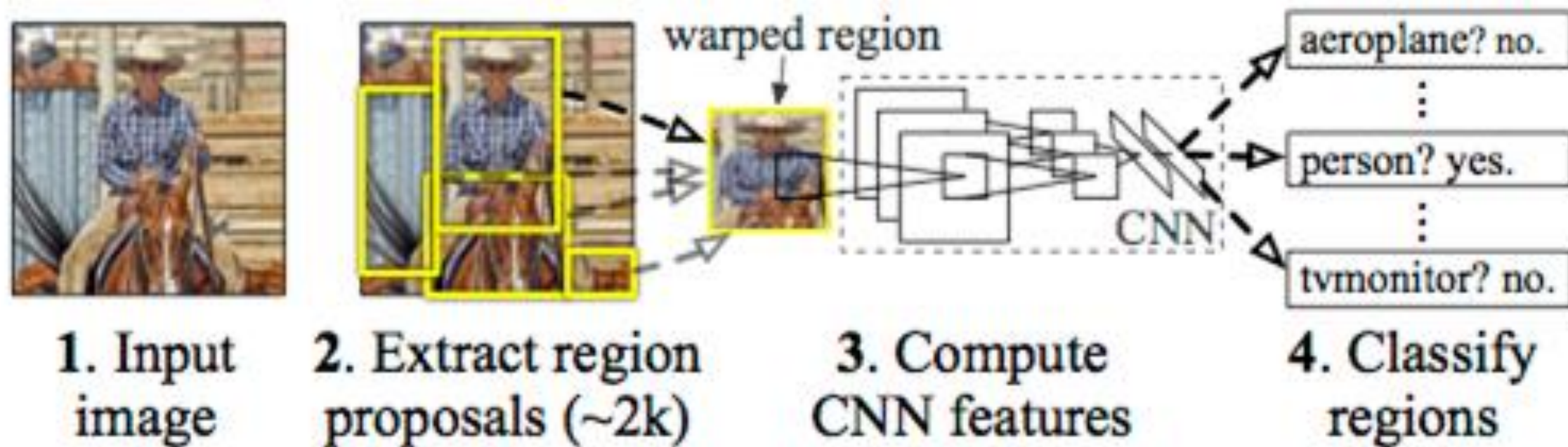
Convnets perform classification



“tabby
cat”

end-to-end learning

R-CNN



R-CNN does detection



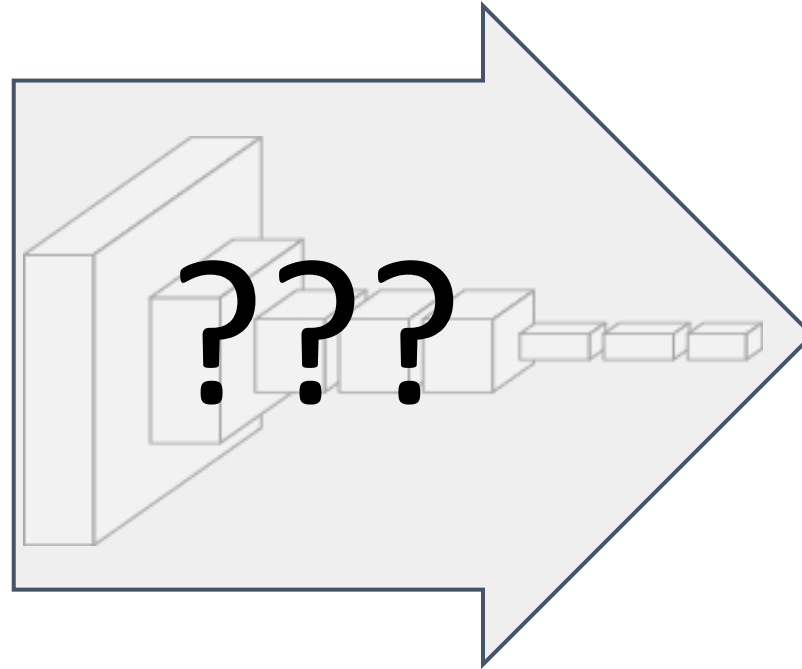
many seconds

R-CNN

“dog”

“cat”

Segmentation?



end-to-end learning

A classification network

convolution

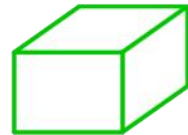
fully connected



$H \times W$



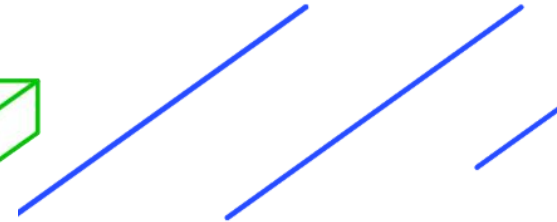
$H/4 \times W/4$



$H/8 \times W/8$



$H/16 \times W/16$



“tabby
cat”

Change to fully convolutional

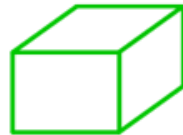
convolution



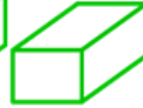
$H \times W$



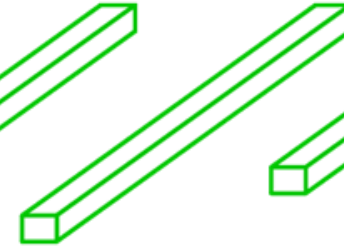
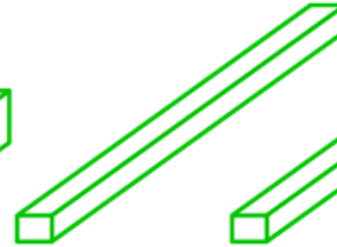
$H/4 \times W/4$



$H/8 \times W/8$

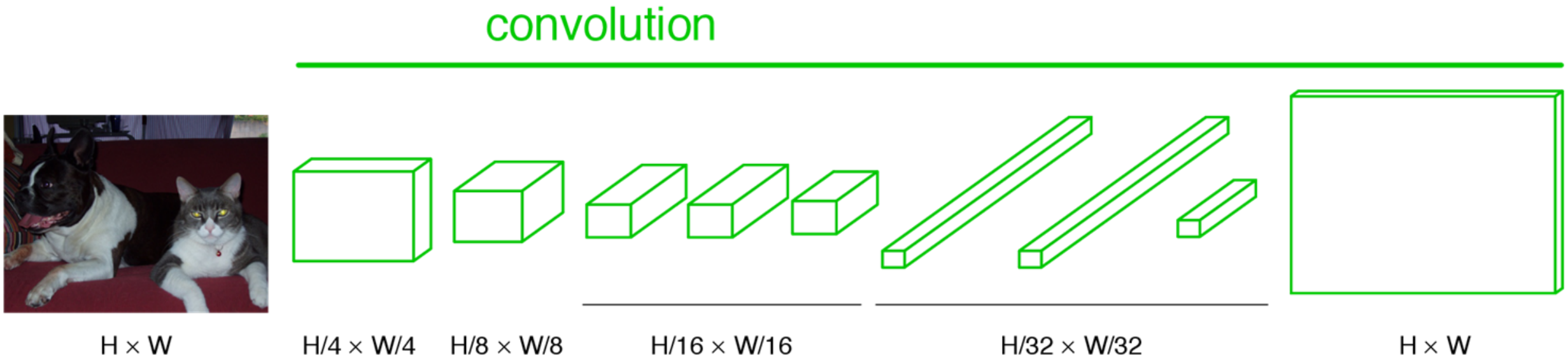


$H/16 \times W/16$



$H/32 \times W/32$

Upsampling output



End-to-end, pixels-to-pixels network

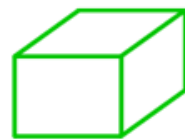
convolution



$H \times W$



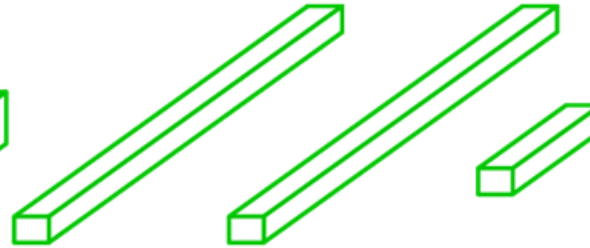
$H/4 \times W/4$



$H/8 \times W/8$



$H/16 \times W/16$



$H/32 \times W/32$



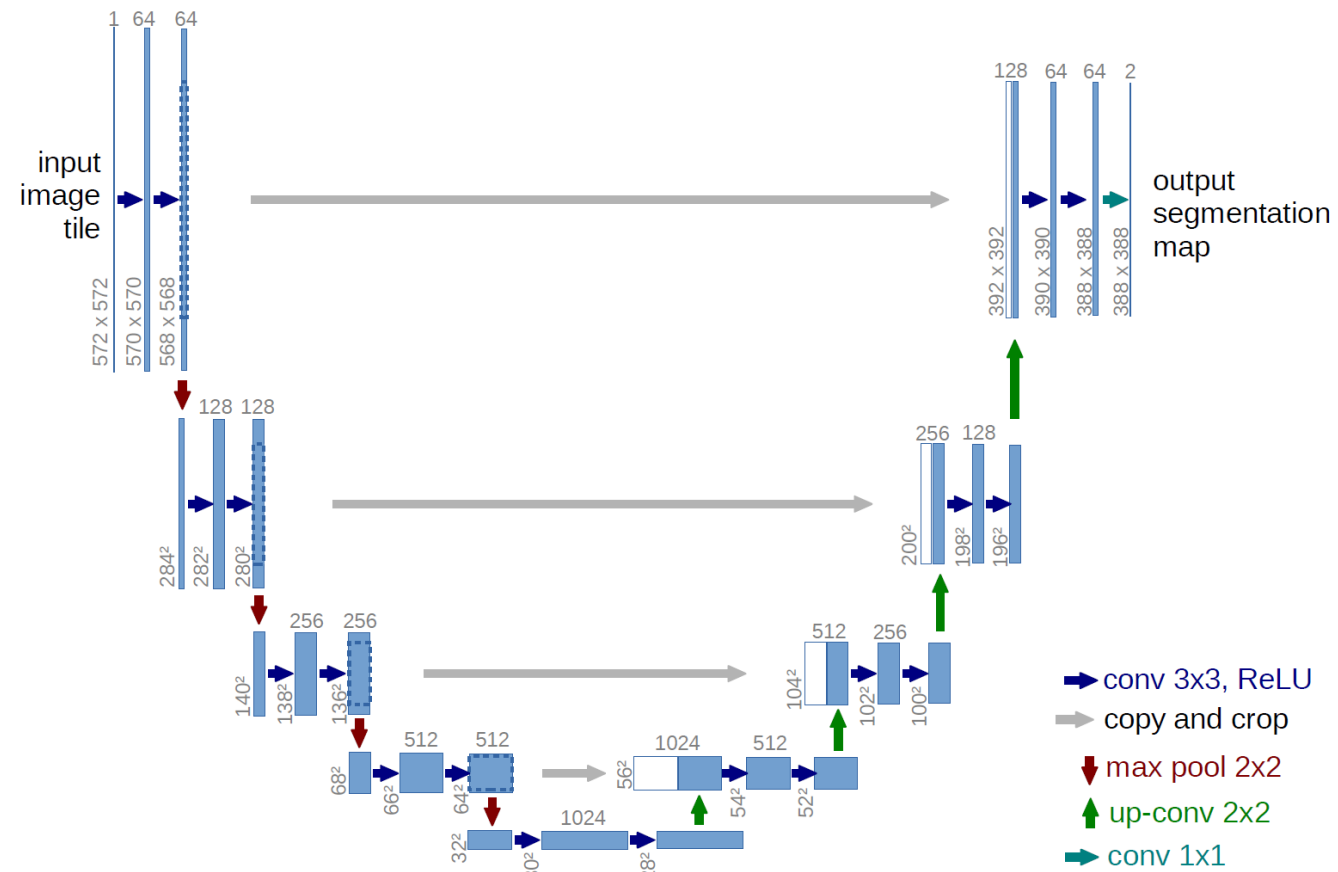
$H \times W$

conv,
pool,
nonlineari
ty

upsampli
ng

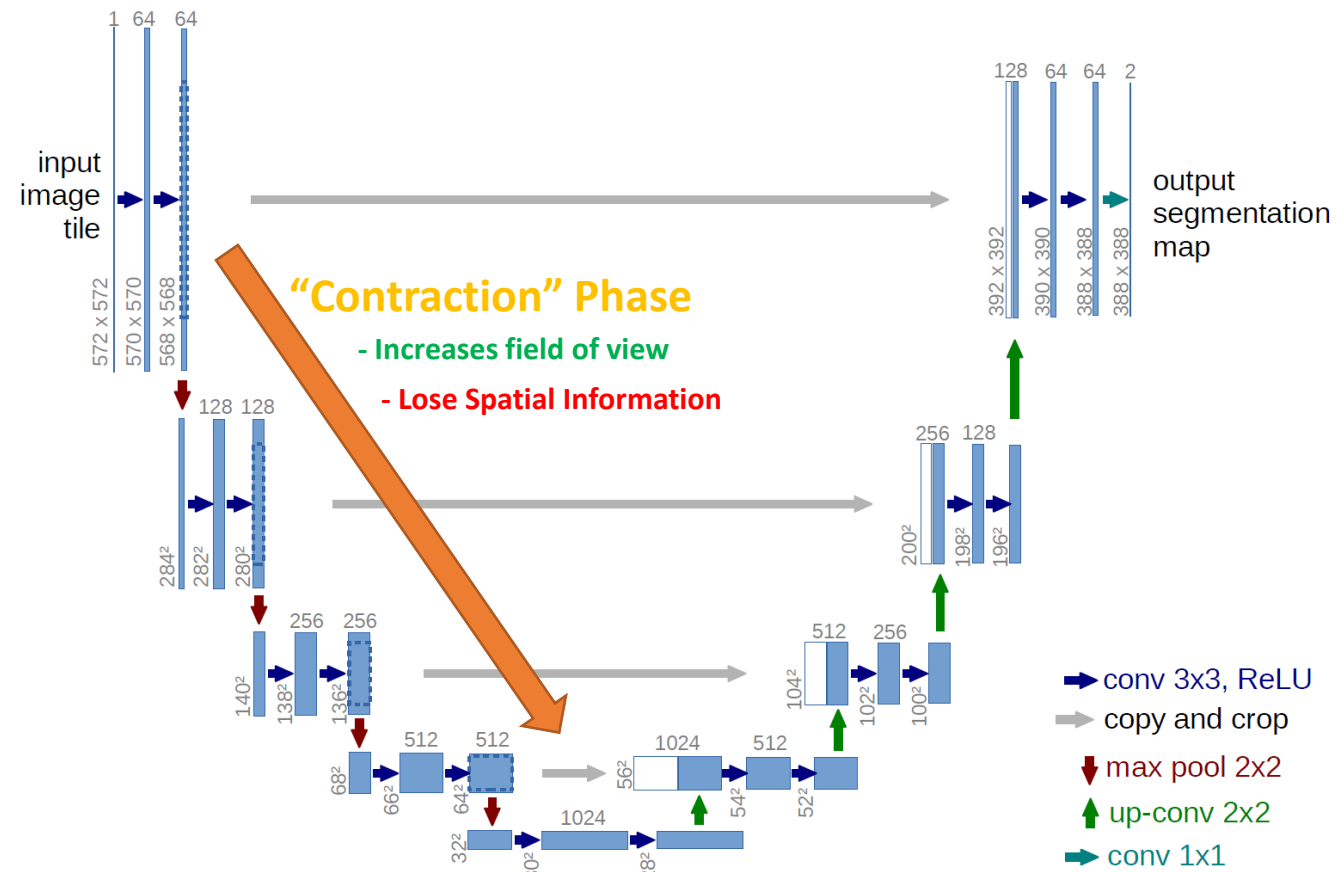
pixelwise
output + loss

U-Net Architecture

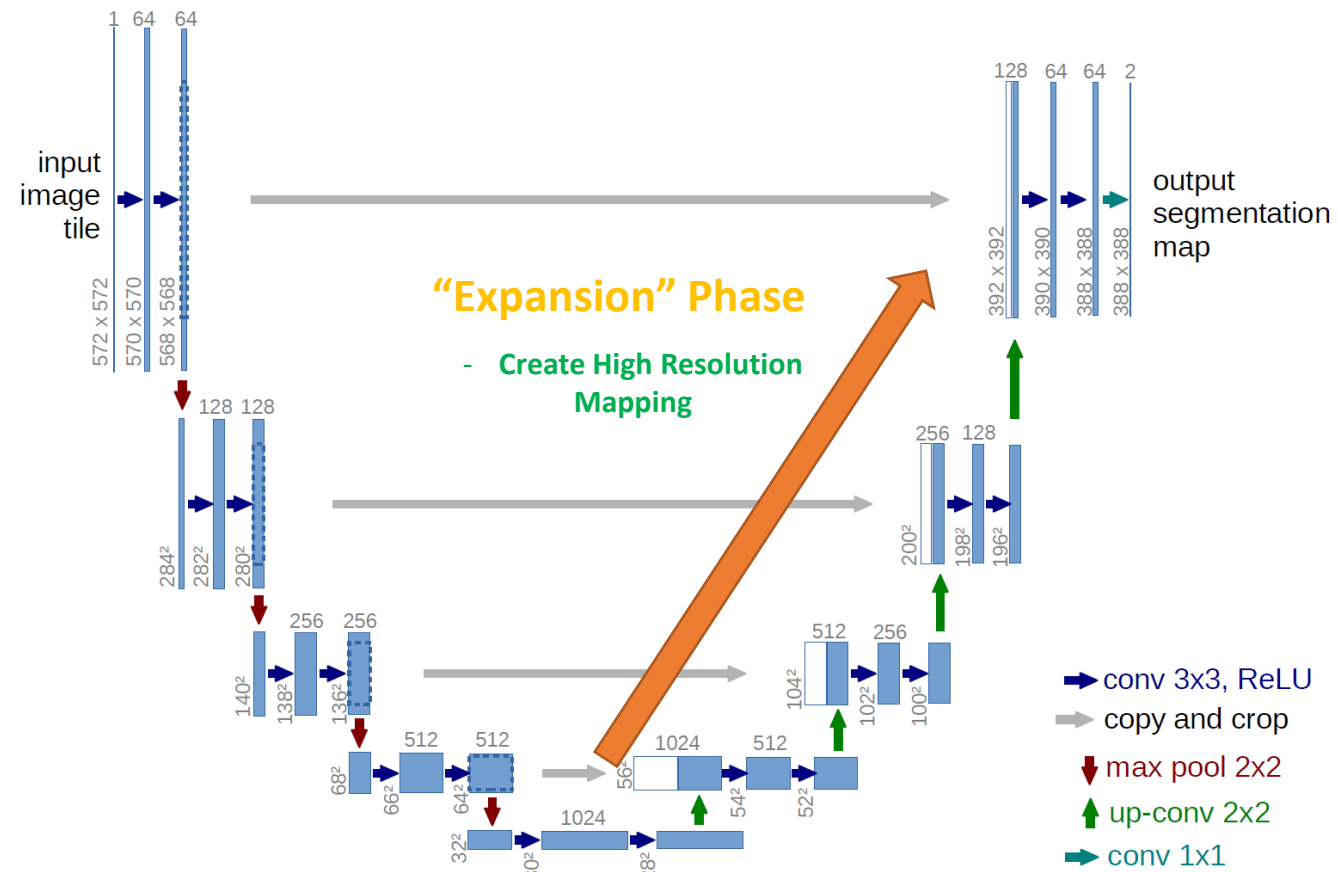


Ronneberger et al. (2015) U-net
Architecture

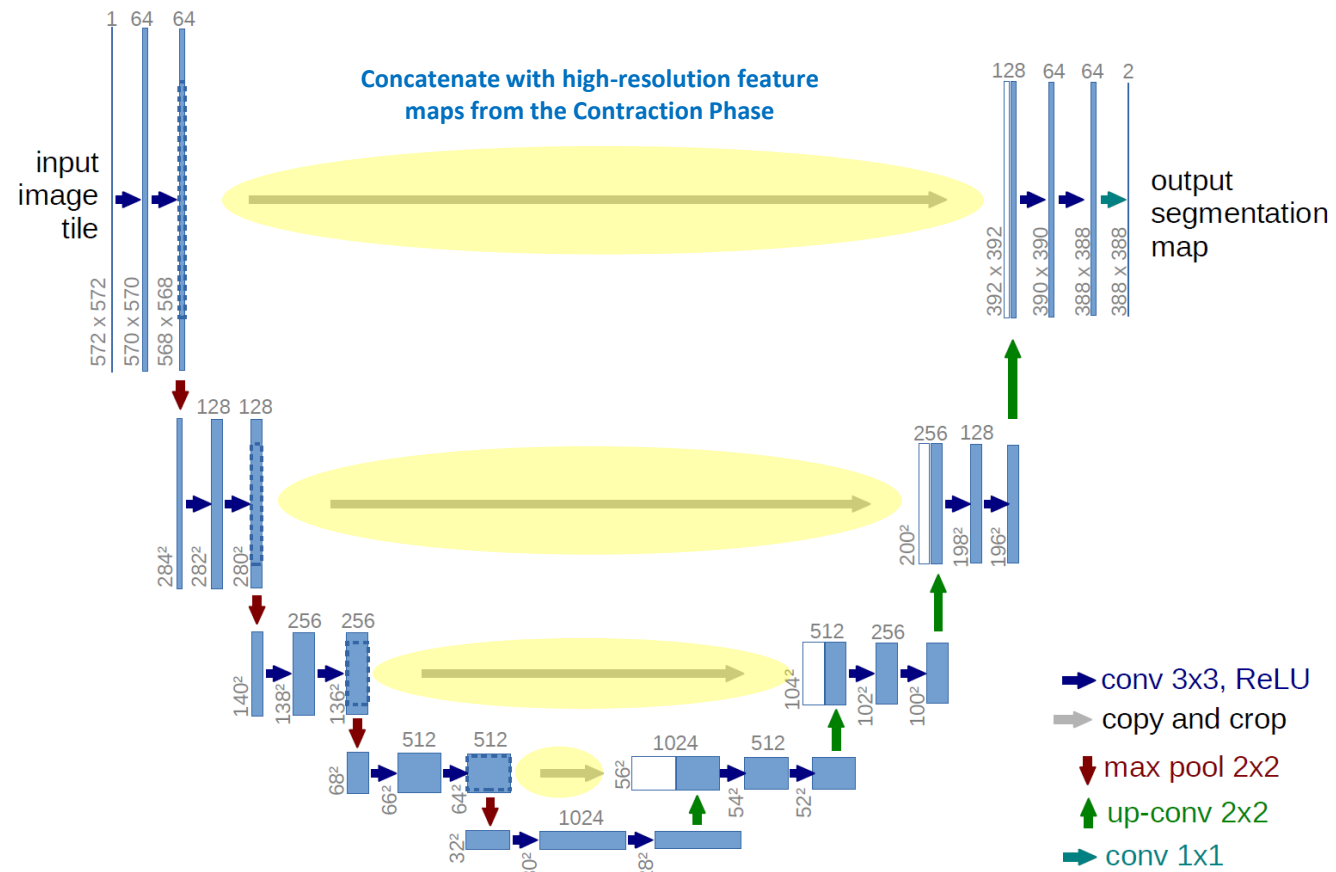
U-Net Architecture



U-Net Architecture



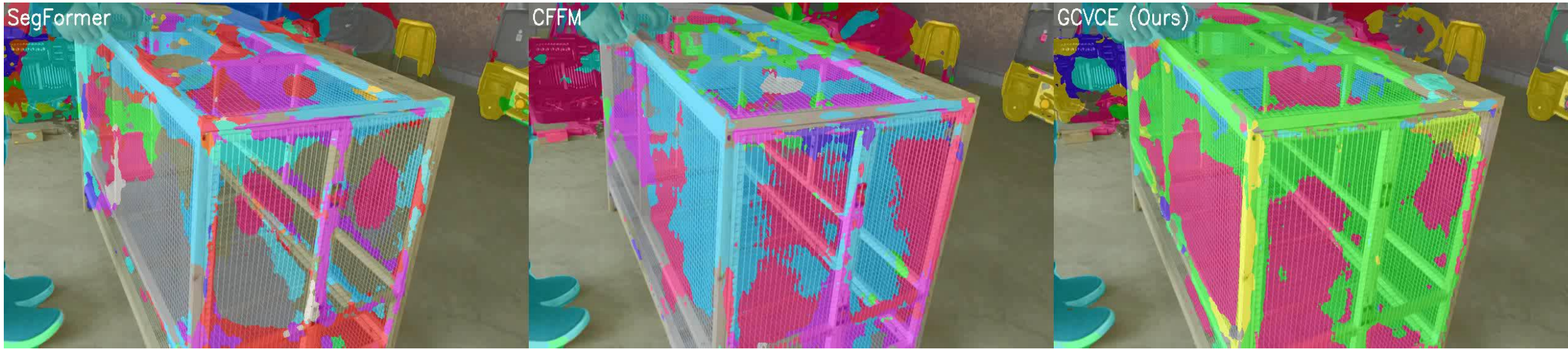
U-Net Architecture



U-Net Summary

- Contraction Phase
 - Reduce spatial dimension, but increases the “what.”
- Expansion Phase
 - Recovers object details and the dimensions, which is the “where.”
- Concatenating feature maps from the Contraction phase helps the Expansion phase with recovering the “where” information.

Image Segmentation vs. Video Segmentation



SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers, NeurIPS 2021

CFFM: Coarse-to-Fine Feature Mining for Video Semantic Segmentation, CVPR 2022

- Image Segmentation
 - Frame based
 - Flicker artefact
- Video Segmentation
 - Frame-frame association
 - Smooth in continuous frames.

<https://www.youtube.com/watch?v=hGrJ3zuuvRQ>

Businesswoman in China caught 'jaywalking'



What is missing?

1. Real human vs photo
2. 2D vs 3D
3. Realistic motion